

Advantage SIS Automation

Setup Guide - Step by Step (Terminal Edition)

How to set up your laptop so a Python program reads your Excel file and enters the data into Advantage SIS for you.

Simple words. Every step uses the **terminal** (type a command, press Enter). Follow the steps in order.

Item	Value used in this guide
Advantage site (DEV)	https://advantage-dev-ui.famecloud.us/login
Project folder	C:\Automation\advantage-sis-bot
Test program name	ZZ Auto Test Shahzaib 01
Excel file name	data\cosmetology.xlsx

What you use	What it does
VS Code	Where you type commands and edit files (already installed)
Python	The programming language
Playwright	Opens the browser and clicks buttons for you
pandas + openpyxl	Read your Excel file
python-dotenv	Keeps your username and password in a private file

How it works: Excel file → Python reads each row → Playwright opens Advantage → logs in → opens the program → types the data → saves → writes a log → next row.

What is in this guide

Part	Topic
Part 1	Install everything (Steps 1 - 5)
Part 2	Set up your project files (Steps 6 - 7)
Part 3	Record one full entry and save it (Step 8)
Part 4	Read the Excel file (Step 9)
Part 5	Build the full automation (Step 10)
Part 6	Test safely, a little at a time (Step 11)
Part 7	What to send to Claude
Part 8	All terminal commands on one page
Part 9	Common problems and fixes
Part 10	Final checklist

Words you will see

Word	Simple meaning
Terminal	The box at the bottom of VS Code where you type commands. Open it with Terminal → New Terminal (or press Ctrl + `).
Command	A line of text you type in the terminal, then press Enter .
Virtual environment (.venv)	A private box for this project's libraries.
Library	Ready-made code you install with pip, like playwright or pandas.
Codegen	A Playwright tool that watches your clicks and writes the code for you.
code <file>	Terminal command that opens (or creates) a file in VS Code, for example code <code>.env</code>

How to paste into the terminal: right-click in the terminal, or press **Ctrl + V**. Type one command at a time and press Enter after each.

Part 1 - Install everything

Step 1: Install Python

Open VS Code. Open the terminal: **Terminal** → **New Terminal**. Type:

```
python --version
```

- If you see **Python 3.10** or higher: Python is ready. Go to Step 2.
- If you see an error, or the Microsoft Store opens: install Python.

Install from the terminal (easiest):

```
winget install -e --id Python.Python.3.12
```

Or install by hand: go to python.org → Downloads → run the installer → on the first screen tick "**Add python.exe to PATH**" → Install Now.

After installing, **close VS Code completely and open it again**, then check:

```
python --version  
pip --version
```

Check it worked: both commands show a version number.

Step 2: Install the VS Code extensions (from the terminal)

```
code --install-extension ms-python.python  
code --install-extension GrapeCity.gc-excelviewer
```

The first one is **required** (Python + Pylance). The second one (Excel Viewer) is optional.

Check it worked: `code --list-extensions` shows `ms-python.python`.

If the terminal says '**code**' is not recognized: install the extension by hand instead. Press **Ctrl + Shift + X**, search **Python**, install the one by Microsoft.

Step 3: Create the project folder (from the terminal)

```
mkdir C:\Automation\advantage-sis-bot  
cd C:\Automation\advantage-sis-bot  
mkdir data, logs  
code -r .
```

What each line does:

1. Creates the project folder.
2. Moves the terminal into that folder.
3. Creates two folders inside it: **data** (Excel files) and **logs** (results and screenshots).

4. Opens this folder in VS Code (in the same window).

After VS Code reloads, open the terminal again (**Terminal** → **New Terminal**). It should already be in the project folder.

Check it worked: type `dir`. You should see **data** and **logs**.

At the end, your project will look like this:

```
advantage-sis-bot/
|-- data/           <- your Excel files
|-- logs/           <- results and error screenshots
|-- .venv/          <- private Python box (created in Step 4)
|-- .env            <- your login details (private, never share)
|-- step1_open.py   <- test: does the browser open?
|-- step2_read_excel.py <- test: can Python read the Excel?
|-- recorded.py     <- your recording (Step 8)
|-- main.py         <- the full automation
```

Step 4: Create and turn on the virtual environment

```
python -m venv .venv
.venv\Scripts\activate
```

Now the start of the terminal line should show **(.venv)**.

If you see "running scripts is disabled on this system": type this once, then run the activate command again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
(If it asks a question, type Y and press Enter.)
```

Tell VS Code to use it: press **Ctrl + Shift + P** → type **Python: Select Interpreter** → choose the one with **.venv**.

Check it worked: the terminal line starts with **(.venv)**.

Every time you open VS Code later, first run: `.venv\Scripts\activate`

Step 5: Install the libraries and the browser

With **(.venv)** showing:

```
python -m pip install --upgrade pip
pip install playwright pandas openpyxl python-dotenv
playwright install chromium
```

Library	What it does
playwright	Controls the browser (clicks, typing, saving)
pandas	Reads the Excel data row by row

Library	What it does
openpyxl	The Excel engine pandas uses for .xlsx files
python-dotenv	Loads your login from the .env file

If "**playwright install chromium**" fails (company network), skip it. Use **Microsoft Edge**, which is already on your laptop. In commands add `--channel=msedge`, and in code use `channel="msedge"`.

Save the list of libraries (useful later):

```
pip freeze > requirements.txt
```

Check it worked: `pip list` shows playwright, pandas, openpyxl and python-dotenv.

Part 2 - Set up your project files

Step 6: Save your login in the .env file

Create and open the file from the terminal:

```
code .env
```

Paste this into the file, change the username and password, then press **Ctrl + S**:

```
SIS_URL=https://advantage-dev-ui.famecloud.us/login
SIS_USER=your_username
SIS_PASS=your_password
```

Or create it fully from the terminal (change the values first):

```
Set-Content .env "SIS_URL=https://advantage-dev-ui.famecloud.us/login"
Add-Content .env "SIS_USER=your_username"
Add-Content .env "SIS_PASS=your_password"
```

Check it worked: Get-Content .env shows the three lines.

Keep .env private. Don't email it, don't put it in a shared folder, and don't send your password to anyone (including Claude).

Step 7: First test - does the Advantage page open?

Create the file from the terminal:

```
code step1_open.py
```

Paste this code and press **Ctrl + S**:

```
import os
from dotenv import load_dotenv
from playwright.sync_api import sync_playwright

load_dotenv()

with sync_playwright() as p:
    browser = p.chromium.launch(headless=False, slow_mo=300)
    # If Chromium did not install, use Edge instead:
    # browser = p.chromium.launch(channel="msedge", headless=False, slow_mo=300)
    page = browser.new_page()
    page.goto(os.getenv("SIS_URL"))
    page.pause() # stops here and opens the Playwright Inspector
    browser.close()
```

Run it from the terminal:

```
python step1_open.py
```

What you should see: a browser opens on the **Advantage DEV login page**, and a small **Playwright Inspector** window opens. Close the browser to finish.

Check it worked: the Advantage login page opened.

If the page doesn't load: the DEV site may only work on the company network. Connect to the **VPN** and try again.

Part 3 - Record one full entry and save it

You do the steps **by hand one time**. Playwright watches and writes the code into a file for you. You record only **one** subject and **one** workunit. Later the script repeats it for every Excel row.

Step 8a: Get ready

- [] Terminal shows **(.venv)**
- [] You are on the company network / VPN
- [] Test program **ZZ Auto Test Shahzaib 01** exists (or you know which program version to open)

Step 8b: Start recording (saves to a file automatically)

```
playwright codegen --target python -o recorded.py `
https://advantage-dev-ui.famecloud.us/login
```

If Chromium is blocked, use Edge:

```
playwright codegen --channel=msedge --target python -o recorded.py `
https://advantage-dev-ui.famecloud.us/login
```

What the parts mean: **--target python** = write Python code. **-o recorded.py** = save the code into the file recorded.py automatically when you close the browser.

Two windows open: the **browser** (where you click) and the **Playwright Inspector** (shows the code as you click).

Step 8c: In the browser, do this once, slowly

1. **Log in**: type username and password, click Login.
2. Click **Setup**.
3. Click **Academics**.
4. Click **Programs**.
5. Click **Search and Add Programs**.
6. Search for **ZZ Auto Test Shahzaib 01** (or Cosmetology) and click Search.
7. Open the program (the 2nd version if using Cosmetology).
8. **Add 1 subject**: fill **every** field, pick from every dropdown, tick the checkboxes, click **Save**.
9. **Add 1 workunit**: fill **every** field, dropdowns and checkboxes, click **Save**.
10. Stop. Don't click anything else.

Click slowly and wait for each page to load. If you click something wrong, keep going and write down which step it was.

Step 8d: Save and finish

1. **Close the browser window**. Playwright saves the code into **recorded.py** automatically.
2. Check the file exists and look at it:

```
dir recorded.py
code recorded.py
```

Remove your real password and make a safe copy to share (change *YourRealPassword* to your real password in this command):

```
(Get-Content recorded.py) -replace 'YourRealPassword', 'PASSWORD' |
Set-Content recorded.txt
```

Check the password is gone:

```
Select-String -Path recorded.txt -Pattern 'PASSWORD'
```

Check it worked: **recorded.txt** exists and shows **PASSWORD** instead of your real password.

Share only **recorded.txt**. Don't share recorded.py (it still has your password). You can delete it later with `del recorded.py`.

Take screenshots too (optional but helpful)

While recording, press **Windows + Shift + S** on each screen. Save them in the project folder as:

File name	Screen
01_login.png	Login page
02_home.png	Home page with the Setup menu
03_menu.png	Setup → Academics → Programs menu open
04_search_programs.png	Search and Add Programs page
05_results.png	Search results
06_program_version.png	Program screen after opening it
07_add_subject.png	Add Subject form (all fields)
08_add_workunit.png	Add Workunit form (all fields)
09_after_save.png	What you see after clicking Save

Optional: stop logging in every time

```
playwright codegen --save-storage=auth.json `
https://advantage-dev-ui.famecloud.us/login
```

Log in, then close the browser. The code can reuse it with `browser.new_context(storage_state="auth.json")`. Keep **auth.json** private.

Part 4 - Read the Excel file

Step 9: Copy the Excel in and test it

Copy your Excel into the **data** folder from the terminal (change the first path to where your file is):

```
Copy-Item "C:\Users\YourName\Downloads\YourExcelFile.xlsx" data\cosmetology.xlsx
dir data
```

Create the test file:

```
code step2_read_excel.py
```

Paste this and press **Ctrl + S**:

```
import pandas as pd

df = pd.read_excel("data/cosmetology.xlsx", sheet_name=0)

print("Sheets:", list(pd.read_excel("data/cosmetology.xlsx", sheet_name=None).keys()))
print("Columns:", df.columns.tolist())
print(df.head(10))
print("Total rows:", len(df))
```

Run it:

```
python step2_read_excel.py
```

What you should see: sheet names, column names, the first 10 rows, and the total rows.

Close the Excel file in Excel before running, or you may get a "Permission denied" error.

Make the Excel match the Advantage form

Do this **after** recording, so the columns match the real form fields. Rules:

Rule	Example
One column = one field on the Advantage form	Column "Workunit Name" → field "Workunit Name"
One row = one record (one workunit)	Row 5 = Ch 01 Chapter Exam
Dropdown values spelled exactly like Advantage	Written Exam (not written exam)
Checkboxes always Yes / No	Not Y, yes or TRUE
Row 1 is the header	No title rows above it
No merged cells, no empty rows	One clean table

Program	Subject	Subj Wt	Min Grade	Workunit Name	Type	Req.	Must Pass
ZZ Auto Test Shahzaib 01	Written Exams	1	75	Ch 01 Ancient Cultures Quiz	Written Exam	Yes	Yes
ZZ Auto Test Shahzaib 01	Written Exams	1	75	Ch 01 Career Paths Quiz	Written Exam	Yes	Yes

Claude can convert your original Excel into this format. Just send the original file and recorded.txt.

Part 5 - Build the full automation

Step 10: Create main.py

```
code main.py
```

It has two safety switches at the top:

Setting	What it means
LIMIT = 3	Only do the first 3 rows. Increase later.
DRY_RUN = True	Fill the fields but do NOT click Save. Nothing changes in Advantage.

Paste this and press **Ctrl + S**:

```

import os, csv, datetime
import pandas as pd
from dotenv import load_dotenv
from playwright.sync_api import sync_playwright

load_dotenv()
LIMIT = 3          # start with 3 rows, increase later
DRY_RUN = True     # True = fill fields but do NOT click Save

df = pd.read_excel("data/cosmetology.xlsx").head(LIMIT)
os.makedirs("logs", exist_ok=True)
log_file = f"logs/run_{datetime.datetime.now():%Y%m%d_%H%M%S}.csv"

def login_and_navigate(page):
    page.goto(os.getenv("SIS_URL"))
    # >>> paste your recorded steps (login -> open program) here <<<
    # page.get_by_label("Username").fill(os.getenv("SIS_USER"))
    # page.get_by_label("Password").fill(os.getenv("SIS_PASS"))
    # page.get_by_role("button", name="Login").click()
    # ...

def process_row(page, row):
    # >>> paste recorded steps for adding ONE workunit here,
    #     and use Excel values instead of fixed text <<<
    # page.get_by_label("Workunit Name").fill(str(row["Workunit Name"]))
    if not DRY_RUN:
        pass # page.get_by_role("button", name="Save").click()

with sync_playwright() as p, open(log_file, "w", newline="") as f:
    log = csv.writer(f)
    log.writerow(["row", "workunit", "status", "message"])
    browser = p.chromium.launch(headless=False, slow_mo=200)
    # Edge: p.chromium.launch(channel="msedge", headless=False, slow_mo=200)
    page = browser.new_page()
    login_and_navigate(page)

    for i, row in df.iterrows():
        name = row.get("Workunit Name")
        try:
            process_row(page, row)
            log.writerow([i, name, "OK", ""])
            print(f"OK    row {i}: {name}")
        except Exception as e:
            page.screenshot(path=f"logs/error_row{i}.png")
            log.writerow([i, name, "ERROR", str(e)[:200]])
            print(f"ERROR row {i}: {e}")
            break # stop on the first error while testing

    browser.close()

```

Run it:

```
python main.py
```

What main.py does, in simple words

1. Loads your login from .env.
2. Reads the Excel (only the first LIMIT rows).
3. Makes a log file in logs with the date and time in its name.
4. Opens the browser, logs in and opens the program.
5. For each Excel row, fills the form.
6. If it works: writes **OK**. If it fails: takes a screenshot, writes **ERROR**, and stops.
7. Closes the browser.

The parts marked >>> **paste here** <<< come from your recording. Claude will fill these in for you after you send recorded.txt.

Part 6 - Test safely, a little at a time

Step 11: Roll it out slowly

Round	Settings in main.py	What to check
1	LIMIT = 1, DRY_RUN = True	Watch the browser. Right fields, right values? Nothing is saved.
2	LIMIT = 1, DRY_RUN = False	1 row is saved. Check it in Advantage.
3	LIMIT = 3	Check all 3 records and the log.
4	LIMIT = 10, then 20	Any ERROR rows? Look at screenshots in logs.
5	LIMIT = 10000 (all rows)	Only after rounds 1 - 4 worked.

To change a setting, open the file, edit the number, and save:

```
code main.py
```

Always test on the DEV site first (<https://advantage-dev-ui.famecloud.us/login>). Never test on a live school database.

Read the log from the terminal

```
dir logs
$last = Get-ChildItem logs\run_*.csv | Sort-Object LastWriteTime | Select-Object -Last 1
Get-Content $last
```

The second command shows the newest log. Example:

```
row,workunit,status,message
0,Ch 01 Ancient Cultures Quiz,OK,
1,Ch 01 Career Paths Quiz,OK,
2,Ch 01 Chapter Exam,ERROR,Timeout 30000ms exceeded waiting for "Save"
```

If you see ERROR, open the screenshot:

```
start logs\error_row2.png
```

Part 7 - What to send to Claude

#	What	Why
1	Your original Excel file (.xlsx)	To see the columns and convert it
2	recorded.txt (password removed)	To get the exact buttons and fields
3	Screenshots 01 - 09 (optional)	To understand every screen
4	The answers below	To know how it should behave

Answers to fill in

1. Advantage URL: `https://advantage-dev-ui.famecloud.us/login`
2. Browser: Chrome / Edge Did "playwright install chromium" work? yes / no
3. Which Excel sheet(s) to use:
4. One Excel row means: one workunit / one subject / other
5. Program to open: ZZ Auto Test Shahzaib 01 / 2nd Cosmetology version / other: ____
6. Do the subjects already exist? yes / no / some
7. If a record already exists: skip / update / stop
8. How many rows in total (about):

How to send

- **Best:** in the Claude desktop app click **Add folder** and choose `C:\Automation\advantage-sis-bot`. Claude reads everything and writes the finished files back into it.
- **Or:** attach the Excel and recorded.txt in the chat.
- **Never send:** your password, .env, recorded.py or auth.json.

Part 8 - All terminal commands on one page

One-time setup

```
python --version
code --install-extension ms-python.python
mkdir C:\Automation\advantage-sis-bot
cd C:\Automation\advantage-sis-bot
mkdir data, logs
code -r .
python -m venv .env
.venv\Scripts\activate
pip install playwright pandas openpyxl python-dotenv
playwright install chromium
code .env
code step1_open.py
python step1_open.py
```

Record

```
playwright codegen --target python -o recorded.py `
    https://advantage-dev-ui.famecloud.us/login
(Get-Content recorded.py) -replace 'YourRealPassword','PASSWORD' |
    Set-Content recorded.txt
```

Excel

```
Copy-Item "C:\path\to\YourExcelFile.xlsx" data\cosmetology.xlsx
python step2_read_excel.py
```

Every day

```
cd C:\Automation\advantage-sis-bot
.venv\Scripts\activate
python main.py
dir logs
```

Stop a running script

Click in the terminal and press **Ctrl + C**.

Part 9 - Common problems and fixes

Problem	Fix
'python' is not recognized, or Microsoft Store opens	Install Python (Step 1) with "Add to PATH". Restart VS Code.
'code' is not recognized	Create files with File → New File in VS Code instead, and install extensions with Ctrl + Shift + X.
'playwright' is not recognized	The .venv is not active. Run .venv\Scripts\activate. Or use <code>python -m playwright ...</code>
running scripts is disabled on this system	<code>Set-ExecutionPolicy -Scope CurrentUser RemoteSigned</code> , then activate again.
No module named 'playwright' (or pandas)	Activate .venv, then run the pip install command again.
playwright install chromium fails	Use Edge: add <code>--channel=msedge</code> to commands, <code>channel="msedge"</code> in code.
Advantage page doesn't load / times out	Connect to the company VPN. Check SIS_URL in .env has no spaces or quotes.
recorded.py is empty or missing	You must close the browser for it to save. Record again.
FileNotFoundError: data/cosmetology.xlsx	The Excel is not in data, or the name is different. Check with <code>dir data</code> .
PermissionError on the Excel file	Close the file in Excel, then run again.
KeyError: 'Workunit Name'	Column name in code doesn't match the Excel. Check with <code>step2_read_excel.py</code> .
Timeout ... exceeded	Button not found or page slow. Open the error screenshot. The name may differ or the page needs more time.

Part 10 - Final checklist

Installation

- [] python --version shows 3.10 or higher
- [] Python extension installed in VS Code
- [] Project folder C:\Automation\advantage-sis-bot with data and logs
- [] (.venv) shows in the terminal
- [] pip list shows playwright, pandas, openpyxl, python-dotenv
- [] playwright install chromium worked (or you will use Edge)

Project

- [] .env has the DEV URL, username and password
- [] python step1_open.py opens the Advantage DEV login page
- [] recorded.py created by codegen (1 subject + 1 workunit)
- [] recorded.txt created with the password removed
- [] Excel copied to data\cosmetology.xlsx
- [] python step2_read_excel.py shows the columns and rows

Testing

- [] Round 1: 1 row, DRY_RUN = True
- [] Round 2: 1 row saved and checked in Advantage
- [] Round 3: 3 rows
- [] Round 4: 10 - 20 rows, log checked
- [] Round 5: all rows (DEV site first)

Your first task: Steps 1 - 7 until the Advantage DEV login page opens. Then Step 8: record one full entry. Send **recorded.txt** and your **original Excel** to Claude, and Claude will finish main.py for your real screens.